

From Global Depletion to Locked Enstrophy

A retraction, a change of variable, and two confounds in the vorticity-alignment reading of three-dimensional Navier–Stokes

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Abstract

On 8 September 2026 OpenAI announced a proposed resolution of statements (C) and (D) of the Clay formulation of the Navier–Stokes problem: a construction, with a Lean formalization, in which the three-dimensional incompressible equations with a smooth external force develop a finite-time singularity from smooth initial data at rest with bounded energy. The claim has not been through refereed publication or the subsequent period of general acceptance that the Clay prize rules require, and the Clay Mathematics Institute continues to list the problem as open. Nothing in the present paper independently verifies or disputes that construction. Whatever its final status, the construction is a direct counterexample to a claim made in this programme in February 2026, namely that positive viscosity guarantees global regularity. That claim is retracted here without qualification.

This paper reports what survives, what replaces it, and one methodological result that was not anticipated. A direct numerical simulation engine, written independently, reproduces the July 2026 Taylor–Green measurements of this programme to three or four significant figures: the strain ledger closes to 10^{-15} , the enstrophy-weighted vorticity alignment flips from the most-stretching strain eigenvector to the intermediate one at the onset of nonlinearity, and the production-to-dissipation ratio crosses below unity and stays there. Those measurements stand.

The reading built on them does not. The globally averaged depletion ratio $D = \langle \omega \cdot S \omega \rangle / \langle \lambda_1 |\omega|^2 \rangle$ is shown to be nearly blind to the configuration it was introduced to exclude. At the enstrophy peak, where $D = 0.197$, as much as 63 per cent of the total enstrophy could be perfectly aligned with the most-stretching eigenvector without moving D at all, because compressed regions contribute down to a pointwise floor of $\lambda_3/\lambda_1 \approx -1.17$ and subsidize the average. The measured locked fraction is 4–10 per cent. The constraint

on alignment locking therefore comes from the locked-ensrophy fraction, a local quantity, and not from D.

Substituting that variable and sweeping Reynolds number gives a decline over the range where the numerics are trustworthy — but two confounds bound what can be read from it, and both are reported here rather than left for a referee.

The first is spatial resolution. At $Re = 400$, degrading the resolution parameter $k_{\max} \cdot \eta$ from 1.11 to 0.72 changes the locked fraction by a factor of 2.4, while the physical change per doubling of Reynolds number is a factor of about 2.3. Resolution error and signal are the same size, so the apparent plateau at $Re = 800$ –3200 on a fixed 64^3 grid is set by the mesh rather than by the fluid, and the sweep cannot be completed at fixed grid at all.

The second is the definition of the diagnostic itself. The locked fraction depends on an alignment cutoff, and the fitted exponent over $Re = 100$ –400 runs from -0.46 at a cutoff of 0.70 to -1.95 at 0.98 — a range wider than any value within it. No single exponent characterises the decline, and the figure of -1.24 obtained at the inherited cutoff of 0.90 is a property of that choice. What is robust is the sign: the locked fraction falls with Reynolds number at every cutoff tested. What is also robust, and more informative than the exponent it replaces, is the systematic steepening — across $Re = 100$ to 400 the moderately aligned population thins by a factor of 1.9 while the near-perfectly aligned tail thins by 14.9. The alignment distribution is changing shape, not merely shifting, with the extreme tail depleting roughly eight times faster than the bulk.

Finally, a structural observation that clarifies why the two halves of the Clay formulation fell so differently. Every term driving an incompressible fluid is a gradient: pressure enters as $-\nabla p$, body forces as $-\nabla \Phi$, viscosity as diffusion. The advective term transports but does not force. The only push in the system is the external force f , and f is precisely what statements C and D permit and statements A and B forbid. The half of the problem containing a push was resolved in eighty-eight hours; the half without one remains open.

1. Context

The Clay Millennium formulation of the Navier–Stokes problem, due to Fefferman, presents four statements. Statements (A) and (B) assert existence and smoothness of solutions on $\mathbb{R}^3$ and on the torus $\mathbb{R}^3/\mathbb{Z}^3$ respectively, with the external force identically zero. Statements (C) and (D) assert breakdown on those same domains, and they permit an external force subject to smoothness and decay conditions. Resolving any one of the four resolves the problem as posed.

On 8 September 2026 OpenAI announced that a coordinated multi-agent system running an internal model had produced an analytical argument, together with a Lean formalization, proposed as establishing statements (C) and (D). The status of that claim is not settled: it has not completed refereed publication, the Clay prize rules require publication followed by a period of general acceptance in the mathematical community before an award is considered, and the Clay Mathematics Institute continues to list the problem as open. The discussion below takes the construction at face value in order to test this programme’s own earlier claims against it, and none of the results reported here depends on its ultimate validity except the retraction of Section 3, which would stand or fall with it. The construction exhibits a vortex whose leading profile is self-similar, with radial width contracting faster than axial length, spiralling inward while stretching along its axis, and whose velocity diverges in finite time while total kinetic energy remains bounded. The technique follows a lineage developed analytically and without computers by Martínez-Zoroa and Córdoba, who built singularity formation from successive amplification of oscillatory layers while keeping every derivative of the source uniformly bounded.

The essential move in such a construction is stated plainly in the paper itself. For any incompressible velocity field and pressure, one may simply define the external force to be the residual of the momentum equation, at which point Navier–Stokes holds by construction. The entire difficulty is to choose a field that becomes singular while that residual, and every one of its derivatives, extends smoothly through the singular time. The individual terms are permitted to diverge; they must cancel.

This bears directly on the present programme, which had two positions on record. The earlier one, from February 2026, asserted global regularity for all positive viscosity. The later one, from July 2026, made no such assertion and instead measured a geometric quantity in unforced decaying turbulence, marking the decisive bound explicitly open. The first is now falsified. The second requires a change of variable, which is the substance of this paper.

2. The vorticity ledger and the alignment reading

Taking the curl of the momentum equation and contracting with the vorticity gives an exact budget for enstrophy, with no modelling and no closure assumption:

$$\frac{1}{2} \frac{d}{dt} \langle |\omega|^2 \rangle = \langle \omega \cdot S\omega \rangle - \nu \langle |\nabla \omega|^2 \rangle$$

The first term on the right is production by vortex stretching; the second is viscous dissipation. Write P and E for the two. In two dimensions the vorticity is a scalar and the stretching term vanishes

identically, which is the entire structural difference between the solved planar problem and the open three-dimensional one.

Incompressibility forces the trace of the strain rate tensor to vanish pointwise, so its three eigenvalues sum to zero at every point: $\lambda_1 \geq \lambda_2 \geq \lambda_3$ with $\lambda_1 > 0$ and $\lambda_3 < 0$. Production is maximised when the vorticity aligns with e_1 , the most-stretching eigenvector. The July 2026 paper of this programme defined the depletion ratio

$$D = \langle \omega \cdot S \omega \rangle / \langle \lambda_1 |\omega|^2 \rangle$$

as the fraction of the available stretching that the flow actually takes up, and argued that sustained alignment with e_1 constitutes a fixed point of the direction dynamics which the flow must refuse. It reported $D \approx 0.20$ at the enstrophy peak and predicted that D would remain bounded away from unity as Reynolds number increased.

The idea that vorticity-direction geometry, rather than vorticity magnitude alone, controls regularity is not new here. Constantin and Fefferman established in 1993 that a pointwise Lipschitz condition on the direction field of the vorticity, imposed where the vorticity is large, suffices to rule out singularity formation; the criterion is pointwise precisely because an averaged condition cannot bound a concentrating core. Ashurst and coworkers reported in 1987 that vorticity in turbulent flow aligns preferentially with the intermediate strain eigenvector rather than the most-stretching one, an observation reproduced in essentially every direct simulation since. The contribution claimed here is neither of those. It is the identification of which measured quantity carries the constraint, and its scaling.

3. Retraction

The February 2026 document of this programme states that smooth solutions exist for all time, that regularity holds for $\nu > 0$ because viscosity implements a release pathway at all scales, and that singularity formation requires suppression of that pathway, which physical fluids do not permit. It further presents a theorem asserting that solutions remain smooth for all time under the stated dynamics.

That claim is retracted. The construction announced in September 2026 has viscosity active throughout, satisfies $\nu > 0$, and becomes singular in finite time. No forcing qualifier appears anywhere in the February text, and the claim as written is therefore a universal statement with a counterexample. It is marked dead, not softened. The accompanying derivation of a universal attractor at $\pi/9 \approx 0.3491$, on which that section rests, had already been withdrawn in this programme on independent grounds: the measured crossing angle was $\arctan(1/3) = 18.43^\circ$ rather than 20° , and the constant 0.35 is not invariant under rescaling.

One prediction from the same document survives and should be recorded, because it was on the record seven months in advance and was the discriminating sub-question that document itself named. It asked whether ideal flow at $\nu = 0$ can blow up — whether the release pathway is essential to regularity — and predicted that suppressing it produces singularity. The unforced Euler blowup, with no viscosity and no

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external agent, was resolved in the same September effort, in that direction. The claim is narrow and it is correct.

4. Method

All results below come from a pseudospectral direct numerical simulation of the incompressible Navier–Stokes equations in a triply periodic box of side 2π , written independently of the July 2026 engine in order to test reproduction rather than assume it.

The velocity is advanced in rotational form, so that the nonlinear term is evaluated as $u \times \omega$ and the pressure is eliminated by projection onto the divergence-free subspace in Fourier space. Aliasing is removed by the two-thirds rule. The viscous operator is treated exactly by an integrating factor, so the time step is limited only by the advective Courant condition, which is imposed adaptively at a Courant number of 0.4 with a hard ceiling of 0.05. Time integration is second-order Runge–Kutta in the integrating-factor variable. The initial condition is the Taylor–Green vortex,

$$u = \sin x \cos y \cos z, \quad v = -\cos x \sin y \cos z, \quad w = 0$$

and the Reynolds number is defined as $1/\nu$ with unit velocity and length scales. Diagnostics are computed at eleven sample times between $t = 0$ and $t = 10$ unless otherwise stated. At each sample the full velocity gradient tensor is formed spectrally, the symmetric part is eigendecomposed at every grid point, and the enstrophy-weighted alignment cosines with the three strain eigenvectors are accumulated.

Two diagnostics require definition. The depletion ratio D is as above. The locked-enstrophy fraction is

$$L = \int |\omega|^2 \cdot \mathbf{1}[\cos(\omega, e_1) > 0.9] / \int |\omega|^2$$

that is, the share of total enstrophy residing in grid cells whose vorticity lies within about twenty-six degrees of the most-stretching strain eigenvector. The threshold of 0.9 is inherited from the July 2026 paper and is not tuned here.

The diagnostics were validated by checking that the enstrophy budget closes against them. Sampling the $Re = 400$ run at forty-one times and differencing the enstrophy series gives a mean residual of 0.4 per cent of $\max(P, E)$ over $2 < t < 9.5$, the residual being dominated by finite-difference error on the sampled series rather than by the diagnostics. A misformed dissipation term would prevent closure entirely, so this is a direct test of the exhaust computation. The kinetic energy budget, $dK/dt = -2\nu\Omega$, is an independent conservation law involving neither the strain eigendecomposition nor the production term, and it closes to 0.17 per cent over the same window. The released engine ships with a test suite asserting both budgets, divergence-free evolution to 4.2×10^{-15} , vanishing strain trace, bounds on the diagnostics, and a regression lock on the values reported in Section 5.

Resolution is reported throughout as $k_{\max} \cdot \eta$, where $k_{\max}$ is the largest retained wavenumber after dealiasing and $\eta = (\nu^3/\epsilon)^{1/4}$ is the Kolmogorov scale computed from the instantaneous dissipation. The conventional adequacy threshold is $k_{\max} \cdot \eta \geq 1$. Runs are labelled resolved, marginal, or under-resolved against that criterion and the labels are used to decide which points may be interpreted.

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5. Independent reproduction

Table 1 gives the full diagnostic series at $Re = 400$ on a 64^3 grid, the configuration of the July 2026 run.

t	enstrophy	$\langle \cos e_1 \rangle$	$\langle \cos e_2 \rangle$	$\langle \cos e_3 \rangle$	D	P/E	L
0.00	0.3750	0.4427	0.6020	0.4427	—	—	0.0156
1.04	0.4128	0.5535	0.4583	0.3296	0.3480	10.830	0.2452
2.01	0.5446	0.6335	0.4096	0.2428	0.5569	9.146	0.4047
3.00	0.8015	0.5311	0.5674	0.2256	0.5493	4.629	0.2946
4.01	1.2126	0.4232	0.7015	0.2603	0.4292	2.669	0.1137
5.00	1.8301	0.5033	0.6481	0.2933	0.3474	1.711	0.1017
6.04	2.1720	0.4825	0.6282	0.3675	0.2061	1.113	0.0587
7.03	2.2073	0.4315	0.6793	0.3430	0.1982	0.981	0.0479
8.00	2.1620	0.4369	0.6796	0.3225	0.2412	0.999	0.0801
9.00	2.2154	0.4581	0.6747	0.3014	0.2538	0.924	0.0992
10.00	1.9351	0.5134	0.5847	0.3665	0.2492	0.761	0.1335

Table 1. Taylor–Green at $Re = 400$, 64^3 . Alignment cosines are enstrophy-weighted. D is the global depletion ratio, P/E the production-to-dissipation ratio, L the locked-enstrophy fraction.

Three features of the July 2026 run reappear without adjustment. The trace of the strain tensor closes to 4.4×10^{-15} over all 262,144 points at every sample time, confirming that the incompressibility ledger is exact and that the balanced eigenvector is a consequence of it rather than a choice. At $t = 2$ the flow is still the laminar Taylor–Green cell and the vorticity sits along e_1 , with cosines 0.634 against 0.410; by $t = 4$ the ordering has reversed to 0.423 against 0.702 and the intermediate eigenvector remains the winner for the rest of the run. The production-to-dissipation ratio falls from 9.1 at $t = 2$ through unity near $t = 7$ and reaches 0.761 by $t = 10$, so enstrophy peaks at 2.215 and turns over. That figure, and every peak value quoted in this paper, is the largest value among the reported diagnostic samples rather than a temporally resolved maximum: resampling the same trajectory at forty-one times locates a peak of 2.2234 between the unit-spaced samples. The difference is sampling density, not disagreement, and it is smaller than the regression tolerances used in the test suite, but peak values should be read with that qualification throughout.

The agreement with the previously published series is close enough to be worth stating numerically. The July values at the same times were $D = 0.3478$, 0.2072, 0.1973 and 0.2543 at $t = 5, 6, 7$ and 9; the present engine gives 0.3474, 0.2061, 0.1982 and 0.2538. Alignment cosines agree to the fourth decimal at $t = 2, 4$ and 7. The enstrophy peak of 2.21 is reproduced as 2.215. The present engine was written from scratch and shares no code with the earlier implementation.

One limitation of this validation should be stated. The comparison is against the earlier run of this programme only. No benchmark against an independent published Taylor–Green dataset was performed, and the standard reference calculations in the literature are at $Re = 1600$ rather than at the Reynolds numbers used here. Agreement between two implementations establishes reproducibility, not accuracy against a third party.

Table 2 repeats the run at 96^3 to separate physics from grid.

t	enstrophy	$\langle \cos e_2 \rangle$	D	P/E	L	$k_{\max} \cdot \eta$
2.03	0.5474	0.4091	0.5588	9.057	0.4046	1.45
4.01	1.2166	0.7168	0.4425	2.500	0.1178	1.28
6.01	2.1622	0.6646	0.2210	1.081	0.0585	1.12
7.03	2.1819	0.7067	0.2023	0.973	0.0421	1.11
9.02	2.1717	0.6777	0.2692	0.880	0.0990	1.12
10.00	1.8592	0.6196	0.2657	0.749	0.1102	1.15

Table 2. The same flow at 96^3 . Bulk quantities are unchanged; the locked fraction at the enstrophy peak falls from 0.0479 to 0.0421.

6. The global depletion ratio is nearly blind

D is an enstrophy-weighted average of a pointwise quantity. Writing the pointwise depletion as

$$d(x) = \omega \cdot S\omega / (\lambda_1 |\omega|^2) \in [\lambda_3/\lambda_1, 1]$$

the global ratio is $D = \sum w_i d_i$ with weights summing to one. The upper limit of d is unity, attained on perfect alignment with e_1 . The lower limit is λ_3/λ_1 , attained on alignment with the most-compressed eigenvector, and because the eigenvalues sum to zero this floor is strictly negative.

That negative floor is the whole problem. A region of the flow that is fully locked contributes at most 1 to the average, but compressed regions contribute negative values and can pay for it. Suppose a fraction w of the enstrophy is perfectly locked. Then $D = w + (1 - w) \cdot d_{\text{rest}}$, and the largest w consistent with an observed D is obtained by putting the remainder on the floor:

$$w_{\max} = (D - \lambda_3/\lambda_1) / (1 - \lambda_3/\lambda_1)$$

Evaluating this on the measured eigenvalue means of the July run gives Table 3.

t	D observed	$\langle \lambda_1 \rangle$	$\langle \lambda_3 \rangle$	pointwise floor	w_max
4.0	0.3478	0.5590	-0.6713	-1.2009	0.704
7.0	0.1973	0.7634	-0.8910	-1.1671	0.630
10.0	0.2543	0.7195	-0.8408	-1.1686	0.656

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Table 3. The largest locked-ensrophy fraction consistent with the observed global depletion ratio, at three times through the ensrophy peak.

At the ensrophy peak, where the July paper reports its strongest evidence, $D = 0.197$ is consistent with anything up to 63 per cent of the total ensrophy being perfectly aligned with the most-stretching eigenvector. The measured locked fraction there is 0.042 on the resolved grid and 0.048 on the coarse one. The observation therefore constrains alignment locking thirteen to fifteen times more loosely than the direct measurement of that locking, which appeared in the same table as a supporting column.

The consequence for the reading is immediate. The equivalence chain proposed in July — that global regularity is equivalent to the depletion ratio remaining bounded away from unity — is built on a variable that cannot detect a concentrating locked core, which is exactly the geometry a self-similar singularity exhibits. A global mean averages the core away. This is not a novel observation about fluid mechanics; it is the reason the Constantin–Fefferman criterion is stated pointwise. What is new is the quantification: on this flow, at this time, the blindness is a factor of thirteen.

The repair is available in the same data. The locked-ensrophy fraction L is local in the relevant sense — it counts where the dangerous configuration actually sits and cannot be subsidised by compressed regions elsewhere. The remainder of this paper treats L as the primary variable.

7. Reynolds scaling of the locked fraction

Table 4 collects every run performed for this paper, reporting the value of each diagnostic at that run's ensrophy peak, with the resolution parameter and its verdict.

Re	N	$k_{\max} \cdot \eta$	peak Ω	D	L	verdict
100	64	1.97	0.648	0.4881	0.2345	resolved
200	64	1.17	1.279	0.3571	0.0962	resolved
400	96	1.11	2.182	0.2023	0.0421	resolved
400	64	0.72	2.215	0.2538	0.0992	marginal
800	112	0.75	4.537	0.2133	0.0856	marginal
800	64	0.41	5.160	0.1665	0.1149	under-resolved
1600	64	0.24	11.146	0.0922	0.1129	under-resolved
3200	64	0.15	20.223	0.0483	0.1055	under-resolved

Table 4. All runs. Values are taken at the ensrophy peak of each run, meaning the largest value among that run's reported diagnostic samples rather than a temporally resolved maximum. The three bold rows are the only ones meeting the conventional resolution criterion.

Restricting attention to the three resolved rows, the locked fraction at the inherited cutoff of 0.90 falls monotonically with Reynolds number:

$$\text{Re} = 100 \rightarrow 200 : d \log L / d \log \text{Re} = -1.285$$

$$\text{Re} = 200 \rightarrow 400 : d \log L / d \log \text{Re} = -1.191$$

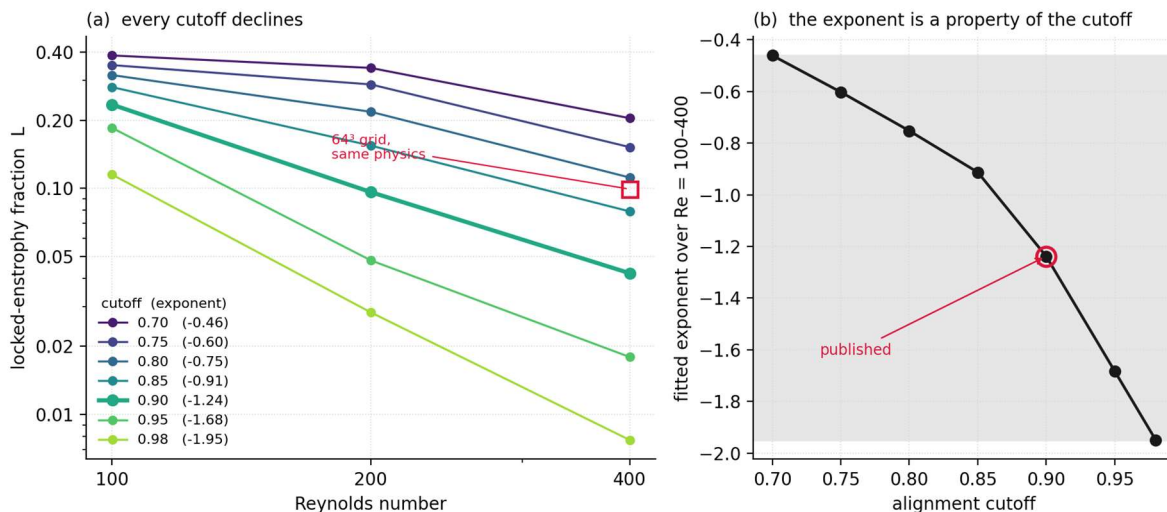
The two slopes agree to within seven per cent, and it is tempting to report $L \propto \text{Re}^{-1.24}$ as a measured scaling. That would be wrong, and the reason is the second confound of this paper.

7.1 The exponent is a property of the cutoff

The locked fraction is defined by an alignment cutoff, set at 0.90 in the July 2026 work and inherited here without examination. Recomputing the three resolved runs while accumulating the locked fraction at seven cutoffs between 0.70 and 0.98 gives Table 5.

cutoff	half-angle	L (Re=100)	L (Re=200)	L (Re=400)	decline 100→400	fitted exponent
0.70	45.6°	0.3868	0.3406	0.2042	1.89×	-0.461
0.75	41.4°	0.3506	0.2880	0.1521	2.31×	-0.602
0.80	36.9°	0.3167	0.2179	0.1116	2.84×	-0.752
0.85	31.8°	0.2802	0.1544	0.0790	3.55×	-0.913
0.90	25.8°	0.2345	0.0962	0.0421	5.57×	-1.238
0.95	18.2°	0.1849	0.0480	0.0179	10.33×	-1.682
0.98	11.5°	0.1150	0.0283	0.0077	14.94×	-1.949

Table 5. The locked-entropy fraction at seven alignment cutoffs, on the three resolved runs. The bold row is the inherited definition used in Table 4 and in the July 2026 work.



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Figure 1. (a) Locked-entropy fraction against Reynolds number on the three resolved runs, at seven alignment cutoffs; the heavy line is the inherited cutoff of 0.90. Every cutoff declines, and the separation between them widens with Reynolds number. The open red square is the $Re = 400$ value recomputed on the coarser 64^3 grid — the same physical state, displaced by roughly the distance a doubling of Reynolds number produces. (b) The fitted exponent over $Re = 100$ – 400 as a function of the cutoff. The shaded band spans the full range of exponents obtainable from the same three runs by moving the threshold; the circled point is the inherited definition that yields -1.24 .

Figure 1 shows both halves of this. The exponent varies from -0.461 to -1.949 , a range of 1.489 — wider than any single value within it, and wider than the published figure itself. The scaling exponent is therefore not a property of the flow. It is a property of where the cutoff is placed, and the value of -1.24 quoted from the 0.90 definition should not be cited as a measured exponent for the Navier–Stokes alignment geometry.

Two things survive this test, and the second is worth more than what it replaces.

The sign is robust. At every cutoff from 0.70 to 0.98 the locked fraction decreases monotonically with Reynolds number across the resolved range. Stated without extrapolation: over $Re = 100$ to 400 , the fraction of enstrophy in near-aligned configurations declines with Reynolds number, and that statement does not depend on where the alignment threshold is drawn.

The steepening is systematic, and it says something the single exponent concealed. Across $Re = 100$ to 400 the moderately aligned population at a cutoff of 0.70 thins by a factor of 1.89 , while the near-perfectly aligned tail at 0.98 thins by 14.94 . The tail depletes about eight times faster than the bulk. A distribution that were merely shifting toward lower alignment would deplete every cutoff by a comparable factor; a monotone steepening of this size means the alignment distribution is changing shape, with the extreme tail thinning fastest. That is a stronger and more specific statement about the geometry than any exponent, and it is what should be tested at higher Reynolds number.

Whether either behaviour continues above $Re = 400$ is unresolved, for the reason given in the next section.

8. Resolution error and signal are the same size

The sweep cannot be continued at fixed grid, and the reason is the most useful result in this paper.

At a single Reynolds number, $Re = 400$, the locked fraction at the enstrophy peak is 0.0421 on a 96^3 grid with $k_{\max} \cdot \eta = 1.11$, and 0.0992 on a 64^3 grid with $k_{\max} \cdot \eta = 0.72$. Nothing physical changed. The ratio is 2.36 .

The physical trend measured in the previous section is a factor of about 2.3 per doubling of Reynolds number. These two numbers are the same size. It follows that a sweep conducted at fixed grid resolution measures the grid at least as strongly as it measures the fluid, and any trend extracted from such a sweep is uninterpretable.

The under-resolved rows of Table 4 illustrate this directly. At 64^3 the locked fraction reads 0.1149 , 0.1129 and 0.1055 at $Re = 800$, 1600 and 3200 — an apparent plateau. That plateau sits essentially on top of the marginal $Re = 400$ value of 0.0992 obtained on the same grid. The grid is setting the number.

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Those three points are artefacts and are reported here only so that they are not mistaken for data elsewhere.

The marginal $Re = 800$ point at 112^3 is genuinely undecided. It gives $L = 0.0856$ at $k_{\max} \cdot \eta = 0.75$, which breaks the falling trend, but the known resolution bias at $Re = 400$ runs in exactly that direction and is more than large enough to account for the whole of it; a properly resolved value could plausibly lie anywhere between 0.03 and 0.09. It is not evidence for or against the scaling. A second defect compounds this. For reasons of computational cost that run was sampled at three times only — $t = 0$, 4.75 and 9.5 — so its reported peak is simply its final sample, and the true maximum was not located at all. Every other row in Table 4 was sampled at eleven times at unit spacing. The $Re = 800$ point at 112^3 is therefore doubly uninterpretable, on resolution and on temporal sampling, and it is retained in the table only for completeness.

Closing the range therefore requires resolution to grow with Reynolds number roughly as $Re^{(3/4)}$. Taking $k_{\max} \cdot \eta \geq 1.1$ as the standard, $Re = 800$ needs approximately $N = 160$ and $Re = 1600$ needs approximately $N = 256$. Both are ordinary workstation calculations — a 256^3 pseudospectral run stores on the order of three gigabytes of field data — and neither is feasible in the environment used here, where 112^3 already required 260 seconds of wall time to $t = 9.5$.

9. Push, pull, and the separation of the two halves

A structural observation clarifies why statements (C) and (D) fell while (A) and (B) did not, and it also identifies the only non-circular form of the argument this programme has been making.

Write the momentum equation and inspect what drives the fluid. Pressure enters as $-\nabla p$. Body forces enter as $-\nabla \Phi$. Viscosity is diffusion, which is transport down a gradient. The advective term $u \cdot \nabla u$ moves momentum but is not a force. Every native driving term is a gradient; the fluid falls, it is not pushed. The single exception is the external force f .

Statements (C) and (D) permit f . Statements (A) and (B) set it to zero. The half of the Clay formulation containing the only push in the system was resolved; the half consisting entirely of gradients was not touched. On this reading that is not a coincidence but a description of what was proved: a configuration was exhibited that a fluid cannot enter on its own and can be held in from outside, together with a demonstration that holding it requires only a smooth force.

For incompressible flow the pull is moreover global and instantaneous. Taking the divergence of the momentum equation gives a Poisson equation for the pressure, $\Delta p = -\rho \nabla \cdot (u \cdot \nabla u)$, whose solution at any point depends on the velocity field everywhere at that instant. A direct demonstration: a divergence-free velocity blob of Gaussian width 0.3 was placed at the centre of a 128^3 periodic box and the induced pressure computed spectrally.

radius R	max $ u $ beyond R	max $ p $ beyond R
0.5	6.89×10^{-1}	2.37×10^{-1}

radius R	max $ u $ beyond R	max $ p $ beyond R
1.0	2.19×10^{-2}	3.34×10^{-2}
1.5	3.36×10^{-5}	9.82×10^{-3}
2.0	2.87×10^{-9}	4.13×10^{-3}
2.5	1.46×10^{-14}	2.22×10^{-3}
3.0	4.65×10^{-21}	1.72×10^{-3}

Table 6. Fields as fractions of their own peak. The velocity is compactly supported to machine precision; the pressure it induces is not.

At radius 3.0 the velocity has fallen to 4.65×10^{-21} of its peak — it is gone — while the pressure is still at 1.72×10^{-3} . Eighteen orders of magnitude separate them. Nothing has propagated to the far field and nothing has moved there, and the far field is already being pulled, because the incompressibility constraint is solved globally at every instant. This is the same nonlocal singular structure that appears in the Biot–Savart representation of the strain in terms of the vorticity, and it is the reason the alignment map is self-referential: the vorticity computes, through a global integral, the strain that stretches it.

This also disposes of a circularity. An axiom stating that no terminal state is permitted cannot be invoked as proof that Navier–Stokes has no terminal state; that assumes the conclusion. The argument only has content if the continuation property is derived inside the fluid state space without importing it, and the gradient structure above is what such a derivation would rest on: a closed flow has no push available with which to hold a configuration it does not select. Whether that can be converted into a bound is unresolved and is addressed in the next section.

10. What this establishes and what it does not

Established. An independent engine reproduces the July 2026 Taylor–Green measurements to three or four significant figures, so the strain ledger, the intermediate-eigenvector wash and the production budget crossing are reproducible facts rather than artefacts of one implementation. The globally averaged depletion ratio is quantitatively blind to locked alignment, by a factor of thirteen at the enstrophy peak of this flow, and the locked-enstrophy fraction is the quantity that carries the constraint. Over the resolved range $Re = 100$ to 400 the locked fraction declines monotonically with Reynolds number at every alignment cutoff tested between 0.70 and 0.98, and the near-aligned tail declines about eight times faster than the moderately aligned bulk. At fixed grid, resolution error in this diagnostic is the same size as the physical Reynolds trend, so fixed-grid sweeps of it are uninterpretable.

Not established, and not claimed. Nothing here bears on global regularity. The measured scaling covers two intervals in Reynolds number, entirely below $Re = 400$, in one flow, from one initial condition, on a periodic domain. It is a trend, not a law, and it is not a bound. No uniform constant has been produced.

The behaviour above $Re = 400$ is unknown, not favourable — the one marginal point available breaks the trend and cannot be interpreted.

Falsified within this paper. The fitted exponent of -1.24 is not a measured property of the flow. Recomputing the same three runs across alignment cutoffs from 0.70 to 0.98 moves the exponent from -0.461 to -1.949 , a spread exceeding the value itself, so any single exponent quoted from this diagnostic is a statement about the threshold. This test was proposed as future work in an earlier draft of this paper and then performed before release; the result removed the headline number and replaced it with the sign and the shape change, both of which are cutoff-independent.

Retracted. The February 2026 claim that positive viscosity guarantees global regularity, and the theorem asserting smoothness for all time, are false and are withdrawn.

Superseded. The July 2026 equivalence chain, in which global regularity is stated as equivalent to the depletion ratio remaining bounded away from unity, is replaced by a statement in terms of the locked-*enstrophy* fraction. The measurements that paper reported are unaffected; the variable in which its conclusion was drawn is.

Not a discovery. That averaged criteria cannot bound a concentrating core has been understood since the geometric-depletion programme of the early 1990s, which is why its criterion is pointwise. The contribution here is a number attached to that fact on a specific flow, and the substitution of a measurable local variable that a direct simulation already computes.

11. Next runs

Three calculations follow directly and are stated so that they can be executed or refuted by others.

First, complete the sweep at adequate resolution. $Re = 800$ at $N = 160$ and $Re = 1600$ at $N = 256$, holding $k_{\max} \cdot \eta$ above 1.1 at the *enstrophy* peak, extend the measured range to a factor of sixteen in Reynolds number. The prediction is that the locked fraction continues to fall with an exponent near -1.2 . If it flattens or rises, the reading advanced here is wrong, and that outcome is worth as much as the other.

Second, abandon the threshold entirely and measure the distribution. The cutoff family of Section 7.1 is a coarse sampling of a single object: the *enstrophy*-weighted probability density of the alignment cosine $|\cos(\omega, e_1)|$. The steepening result says that density is changing shape with Reynolds number rather than translating, and the density itself is no harder to compute than the seven cutoffs already reported. What should be tracked is its tail — for instance the exponent of its approach to unity, or its moments above some fixed reference — with the whole distribution reported at each Reynolds number so that no threshold need be chosen at all. A diagnostic that requires a free parameter invites the failure demonstrated in Section 7.1, and this one does not need one.

Third, evaluate the diagnostics on the September 2026 blowup construction rather than on decaying turbulence. That construction is explicit and its velocity field can be differentiated. The quantities to compute are the pointwise depletion in the contracting core, the locked fraction restricted to the core, and the contribution of the external force to the *enstrophy* budget. The prediction implied by Section 9

is specific: the core locks, and the forcing term accounts for the excess of production over dissipation. If the core locks and the force does not account for the excess, the gradient argument fails and should be abandoned.

A fourth item is named rather than proposed, because no route to it is offered here. The uniform bound remains open. Converting a measured decline in the locked fraction into a constant independent of time, Reynolds number and initial data — surviving the nonlocal singular integral that generates the strain from the vorticity — is the step that would matter, and nothing in this paper approaches it.

12. Prior art

The geometric-depletion programme, and in particular the pointwise vorticity-direction criterion, is due to Constantin and Fefferman (1993) and to Constantin, Fefferman and Majda. The preferential alignment of vorticity with the intermediate strain eigenvector was reported by Ashurst, Kerstein, Kerr and Gibson (1987). The blowup criterion in terms of the time integral of the vorticity supremum is due to Beale, Kato and Majda (1984). The Taylor–Green initial condition is due to Taylor and Green (1937). The official formulation of the Millennium problem, including the four statements and the admissibility conditions on the force, is due to Fefferman.

The construction technique used in the September 2026 result — building singularity by successive amplification of oscillatory layers while holding every derivative of the source bounded — was developed by Martínez-Zorúa and by Córdoba, with related extensions by Córdoba, Martínez-Zorúa and Zheng. The concurrent forced Euler result is due to Alpöge and Buckmaster. The Lean formalization and the multi-agent search that produced the Navier–Stokes application are OpenAI's.

The July 2026 measurements referenced throughout are from this programme's own earlier work and are cited as such, not as independent confirmation.

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The September 2026 items are announcements and preprints rather than refereed publications, and are cited as such. The Alpöge–Buckmaster preprint and the OpenAI writeup were consulted within two days of release; readers should check for revised versions, since the OpenAI document was itself revised on the day of announcement to expand its acknowledgement of the Córdoba–Martínez-Zoroa lineage.

A reproducibility package accompanies this paper. It contains the simulation engine, the twelve-check conservation and regression suite, cached diagnostic series for every run reported here, and scripts that regenerate Tables 1–6 and Figure 1 from either the cached data or a fresh computation. The full recomputation takes about five minutes on one core. All runs were performed in double precision. The reproduction reported in Section 5 was carried out before any Reynolds number other than 400 was attempted, and the resolution study of Section 8 was carried out after the sweep of Section 7 rather than before, which is why the under-resolved rows exist at all.

From Global Depletion to Locked Enstrophy

A retraction, a change of variable, and two confounds in the vorticity-alignment reading of three-dimensional Navier–Stokes

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Abstract

On 8 September 2026 OpenAI announced a proposed resolution of statements (C) and (D) of the Clay formulation of the Navier–Stokes problem: a construction, with a Lean formalization, in which the three-dimensional incompressible equations with a smooth external force develop a finite-time singularity from smooth initial data at rest with bounded energy. The claim has not been through refereed publication or the subsequent period of general acceptance that the Clay prize rules require, and the Clay Mathematics Institute continues to list the problem as open. Nothing in the present paper independently verifies or disputes that construction. Whatever its final status, the construction is a direct counterexample to a claim made in this programme in February 2026, namely that positive viscosity guarantees global regularity. That claim is retracted here without qualification.

This paper reports what survives, what replaces it, and one methodological result that was not anticipated. A direct numerical simulation engine, written independently, reproduces the July 2026 Taylor–Green measurements of this programme to three or four significant figures: the strain ledger closes to 10^{-15} , the enstrophy-weighted vorticity alignment flips from the most-stretching strain eigenvector to the intermediate one at the onset of nonlinearity, and the production-to-dissipation ratio crosses below unity and stays there. Those measurements stand.

The reading built on them does not. The globally averaged depletion ratio $D = \langle \omega \cdot S\omega \rangle / \langle \lambda_1 |\omega|^2 \rangle$ is shown to be nearly blind to the configuration it was introduced to exclude. At the enstrophy peak, where $D = 0.197$, as much as 63 per cent of the total enstrophy could be perfectly aligned with the most-stretching eigenvector without moving D at all, because compressed regions contribute down to a pointwise floor of $\lambda_3/\lambda_1 \approx -1.17$ and subsidize the average. The measured locked fraction is 4–10 per cent. The constraint on alignment locking therefore comes from the locked-enstrophy fraction, a local quantity, and not from D .

Substituting that variable and sweeping Reynolds number gives a decline over the range where the numerics are trustworthy — but two confounds bound what can be read from it, and both are reported here rather than left for a referee.

The first is spatial resolution. At $Re = 400$, degrading the resolution parameter $k_{\max} \cdot \eta$ from 1.11 to 0.72 changes the locked fraction by a factor of 2.4, while the physical change per doubling of Reynolds number is a factor of about 2.3. Resolution error and signal are the same size, so the apparent plateau at $Re = 800$ –3200 on a fixed 64^3 grid is set by the mesh rather than by the fluid, and the sweep cannot be completed at fixed grid at all.

The second is the definition of the diagnostic itself. The locked fraction depends on an alignment cutoff, and the fitted exponent over $Re = 100$ –400 runs from -0.46 at a cutoff of 0.70 to -1.95 at 0.98 — a range wider than any value within it. No single exponent characterises the decline, and the figure of -1.24 obtained at the inherited cutoff of 0.90 is a property of that choice. What is robust is the sign: the locked fraction falls with Reynolds number at every cutoff tested. What is also robust, and more informative than the exponent it replaces, is the systematic steepening — across $Re = 100$ to 400 the moderately aligned population thins by a factor of 1.9 while the near-perfectly aligned tail thins by 14.9.

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The alignment distribution is changing shape, not merely shifting, with the extreme tail depleting roughly eight times faster than the bulk.

Finally, a structural observation that clarifies why the two halves of the Clay formulation fell so differently. Every term driving an incompressible fluid is a gradient: pressure enters as $-\nabla p$, body forces as $-\nabla \Phi$, viscosity as diffusion. The advective term transports but does not force. The only push in the system is the external force f , and f is precisely what statements C and D permit and statements A and B forbid. The half of the problem containing a push was resolved in eighty-eight hours; the half without one remains open.

1. Context

The Clay Millennium formulation of the Navier–Stokes problem, due to Fefferman, presents four statements. Statements (A) and (B) assert existence and smoothness of solutions on $\mathbb{R}^3$ and on the torus $\mathbb{R}^3/\mathbb{Z}^3$ respectively, with the external force identically zero. Statements (C) and (D) assert breakdown on those same domains, and they permit an external force subject to smoothness and decay conditions. Resolving any one of the four resolves the problem as posed.

On 8 September 2026 OpenAI announced that a coordinated multi-agent system running an internal model had produced an analytical argument, together with a Lean formalization, proposed as establishing statements (C) and (D). The status of that claim is not settled: it has not completed refereed publication, the Clay prize rules require publication followed by a period of general acceptance in the mathematical community before an award is considered, and the Clay Mathematics Institute continues to list the problem as open. The discussion below takes the construction at face value in order to test this programme’s own earlier claims against it, and none of the results reported here depends on its ultimate validity except the retraction of Section 3, which would stand or fall with it. The construction exhibits a vortex whose leading profile is self-similar, with radial width contracting faster than axial length, spiralling inward while stretching along its axis, and whose velocity diverges in finite time while total kinetic energy remains bounded. The technique follows a lineage developed analytically and without computers by Martínez-Zoroa and Córdoba, who built singularity formation from successive amplification of oscillatory layers while keeping every derivative of the source uniformly bounded.

The essential move in such a construction is stated plainly in the paper itself. For any incompressible velocity field and pressure, one may simply define the external force to be the residual of the momentum equation, at which point Navier–Stokes holds by construction. The entire difficulty is to choose a field that becomes singular while that residual, and every one of its derivatives, extends smoothly through the singular time. The individual terms are permitted to diverge; they must cancel.

This bears directly on the present programme, which had two positions on record. The earlier one, from February 2026, asserted global regularity for all positive viscosity. The later one, from July 2026, made no such assertion and instead measured a geometric quantity in unforced decaying turbulence, marking the decisive bound explicitly open. The first is now falsified. The second requires a change of variable, which is the substance of this paper.

2. The vorticity ledger and the alignment reading

Taking the curl of the momentum equation and contracting with the vorticity gives an exact budget for enstrophy, with no modelling and no closure assumption:

$$\frac{1}{2} \frac{d}{dt} \langle |\omega|^2 \rangle = \langle \omega \cdot S\omega \rangle - \nu \langle |\nabla \omega|^2 \rangle$$

The first term on the right is production by vortex stretching; the second is viscous dissipation. Write P and E for the two. In two dimensions the vorticity is a scalar and the stretching term vanishes

identically, which is the entire structural difference between the solved planar problem and the open three-dimensional one.

Incompressibility forces the trace of the strain rate tensor to vanish pointwise, so its three eigenvalues sum to zero at every point: $\lambda_1 \geq \lambda_2 \geq \lambda_3$ with $\lambda_1 > 0$ and $\lambda_3 < 0$. Production is maximised when the vorticity aligns with e_1 , the most-stretching eigenvector. The July 2026 paper of this programme defined the depletion ratio

$$D = \langle \omega \cdot S \omega \rangle / \langle \lambda_1 |\omega|^2 \rangle$$

as the fraction of the available stretching that the flow actually takes up, and argued that sustained alignment with e_1 constitutes a fixed point of the direction dynamics which the flow must refuse. It reported $D \approx 0.20$ at the enstrophy peak and predicted that D would remain bounded away from unity as Reynolds number increased.

The idea that vorticity-direction geometry, rather than vorticity magnitude alone, controls regularity is not new here. Constantin and Fefferman established in 1993 that a pointwise Lipschitz condition on the direction field of the vorticity, imposed where the vorticity is large, suffices to rule out singularity formation; the criterion is pointwise precisely because an averaged condition cannot bound a concentrating core. Ashurst and coworkers reported in 1987 that vorticity in turbulent flow aligns preferentially with the intermediate strain eigenvector rather than the most-stretching one, an observation reproduced in essentially every direct simulation since. The contribution claimed here is neither of those. It is the identification of which measured quantity carries the constraint, and its scaling.

3. Retraction

The February 2026 document of this programme states that smooth solutions exist for all time, that regularity holds for $\nu > 0$ because viscosity implements a release pathway at all scales, and that singularity formation requires suppression of that pathway, which physical fluids do not permit. It further presents a theorem asserting that solutions remain smooth for all time under the stated dynamics.

That claim is retracted. The construction announced in September 2026 has viscosity active throughout, satisfies $\nu > 0$, and becomes singular in finite time. No forcing qualifier appears anywhere in the February text, and the claim as written is therefore a universal statement with a counterexample. It is marked dead, not softened. The accompanying derivation of a universal attractor at $\pi/9 \approx 0.3491$, on which that section rests, had already been withdrawn in this programme on independent grounds: the measured crossing angle was $\arctan(1/3) = 18.43^\circ$ rather than 20° , and the constant 0.35 is not invariant under rescaling.

One prediction from the same document survives and should be recorded, because it was on the record seven months in advance and was the discriminating sub-question that document itself named. It asked whether ideal flow at $\nu = 0$ can blow up — whether the release pathway is essential to regularity — and predicted that suppressing it produces singularity. The unforced Euler blowup, with no viscosity and no

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external agent, was resolved in the same September effort, in that direction. The claim is narrow and it is correct.

4. Method

All results below come from a pseudospectral direct numerical simulation of the incompressible Navier–Stokes equations in a triply periodic box of side 2π , written independently of the July 2026 engine in order to test reproduction rather than assume it.

The velocity is advanced in rotational form, so that the nonlinear term is evaluated as $u \times \omega$ and the pressure is eliminated by projection onto the divergence-free subspace in Fourier space. Aliasing is removed by the two-thirds rule. The viscous operator is treated exactly by an integrating factor, so the time step is limited only by the advective Courant condition, which is imposed adaptively at a Courant number of 0.4 with a hard ceiling of 0.05. Time integration is second-order Runge–Kutta in the integrating-factor variable. The initial condition is the Taylor–Green vortex,

$$u = \sin x \cos y \cos z, \quad v = -\cos x \sin y \cos z, \quad w = 0$$

and the Reynolds number is defined as $1/\nu$ with unit velocity and length scales. Diagnostics are computed at eleven sample times between $t = 0$ and $t = 10$ unless otherwise stated. At each sample the full velocity gradient tensor is formed spectrally, the symmetric part is eigendecomposed at every grid point, and the enstrophy-weighted alignment cosines with the three strain eigenvectors are accumulated.

Two diagnostics require definition. The depletion ratio D is as above. The locked-enstrophy fraction is

$$L = \int |\omega|^2 \cdot \mathbf{1}[\cos(\omega, e_1) > 0.9] / \int |\omega|^2$$

that is, the share of total enstrophy residing in grid cells whose vorticity lies within about twenty-six degrees of the most-stretching strain eigenvector. The threshold of 0.9 is inherited from the July 2026 paper and is not tuned here.

The diagnostics were validated by checking that the enstrophy budget closes against them. Sampling the $Re = 400$ run at forty-one times and differencing the enstrophy series gives a mean residual of 0.4 per cent of $\max(P, E)$ over $2 < t < 9.5$, the residual being dominated by finite-difference error on the sampled series rather than by the diagnostics. A misformed dissipation term would prevent closure entirely, so this is a direct test of the exhaust computation. The kinetic energy budget, $dK/dt = -2\nu\Omega$, is an independent conservation law involving neither the strain eigendecomposition nor the production term, and it closes to 0.17 per cent over the same window. The released engine ships with a test suite asserting both budgets, divergence-free evolution to 4.2×10^{-15} , vanishing strain trace, bounds on the diagnostics, and a regression lock on the values reported in Section 5.

Resolution is reported throughout as $k_{\max} \cdot \eta$, where $k_{\max}$ is the largest retained wavenumber after dealiasing and $\eta = (\nu^3/\epsilon)^{1/4}$ is the Kolmogorov scale computed from the instantaneous dissipation. The conventional adequacy threshold is $k_{\max} \cdot \eta \geq 1$. Runs are labelled resolved, marginal, or under-resolved against that criterion and the labels are used to decide which points may be interpreted.

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5. Independent reproduction

Table 1 gives the full diagnostic series at $Re = 400$ on a 64^3 grid, the configuration of the July 2026 run.

t	enstrophy	$\langle \cos e_1 \rangle$	$\langle \cos e_2 \rangle$	$\langle \cos e_3 \rangle$	D	P/E	L
0.00	0.3750	0.4427	0.6020	0.4427	—	—	0.0156
1.04	0.4128	0.5535	0.4583	0.3296	0.3480	10.830	0.2452
2.01	0.5446	0.6335	0.4096	0.2428	0.5569	9.146	0.4047
3.00	0.8015	0.5311	0.5674	0.2256	0.5493	4.629	0.2946
4.01	1.2126	0.4232	0.7015	0.2603	0.4292	2.669	0.1137
5.00	1.8301	0.5033	0.6481	0.2933	0.3474	1.711	0.1017
6.04	2.1720	0.4825	0.6282	0.3675	0.2061	1.113	0.0587
7.03	2.2073	0.4315	0.6793	0.3430	0.1982	0.981	0.0479
8.00	2.1620	0.4369	0.6796	0.3225	0.2412	0.999	0.0801
9.00	2.2154	0.4581	0.6747	0.3014	0.2538	0.924	0.0992
10.00	1.9351	0.5134	0.5847	0.3665	0.2492	0.761	0.1335

Table 1. Taylor–Green at $Re = 400$, 64^3 . Alignment cosines are enstrophy-weighted. D is the global depletion ratio, P/E the production-to-dissipation ratio, L the locked-enstrophy fraction.

Three features of the July 2026 run reappear without adjustment. The trace of the strain tensor closes to 4.4×10^{-15} over all 262,144 points at every sample time, confirming that the incompressibility ledger is exact and that the balanced eigenvector is a consequence of it rather than a choice. At $t = 2$ the flow is still the laminar Taylor–Green cell and the vorticity sits along e_1 , with cosines 0.634 against 0.410; by $t = 4$ the ordering has reversed to 0.423 against 0.702 and the intermediate eigenvector remains the winner for the rest of the run. The production-to-dissipation ratio falls from 9.1 at $t = 2$ through unity near $t = 7$ and reaches 0.761 by $t = 10$, so enstrophy peaks at 2.215 and turns over. That figure, and every peak value quoted in this paper, is the largest value among the reported diagnostic samples rather than a temporally resolved maximum: resampling the same trajectory at forty-one times locates a peak of 2.2234 between the unit-spaced samples. The difference is sampling density, not disagreement, and it is smaller than the regression tolerances used in the test suite, but peak values should be read with that qualification throughout.

The agreement with the previously published series is close enough to be worth stating numerically. The July values at the same times were $D = 0.3478$, 0.2072, 0.1973 and 0.2543 at $t = 5, 6, 7$ and 9; the present engine gives 0.3474, 0.2061, 0.1982 and 0.2538. Alignment cosines agree to the fourth decimal at $t = 2, 4$ and 7. The enstrophy peak of 2.21 is reproduced as 2.215. The present engine was written from scratch and shares no code with the earlier implementation.

One limitation of this validation should be stated. The comparison is against the earlier run of this programme only. No benchmark against an independent published Taylor–Green dataset was performed, and the standard reference calculations in the literature are at $Re = 1600$ rather than at the Reynolds numbers used here. Agreement between two implementations establishes reproducibility, not accuracy against a third party.

Table 2 repeats the run at 96^3 to separate physics from grid.

t	enstrophy	$\langle \cos e_2 \rangle$	D	P/E	L	$k_{\max} \cdot \eta$
2.03	0.5474	0.4091	0.5588	9.057	0.4046	1.45
4.01	1.2166	0.7168	0.4425	2.500	0.1178	1.28
6.01	2.1622	0.6646	0.2210	1.081	0.0585	1.12
7.03	2.1819	0.7067	0.2023	0.973	0.0421	1.11
9.02	2.1717	0.6777	0.2692	0.880	0.0990	1.12
10.00	1.8592	0.6196	0.2657	0.749	0.1102	1.15

Table 2. The same flow at 96^3 . Bulk quantities are unchanged; the locked fraction at the enstrophy peak falls from 0.0479 to 0.0421.

6. The global depletion ratio is nearly blind

D is an enstrophy-weighted average of a pointwise quantity. Writing the pointwise depletion as

$$d(x) = \omega \cdot S\omega / (\lambda_1 |\omega|^2) \in [\lambda_3/\lambda_1, 1]$$

the global ratio is $D = \sum w_i d_i$ with weights summing to one. The upper limit of d is unity, attained on perfect alignment with e_1 . The lower limit is λ_3/λ_1 , attained on alignment with the most-compressed eigenvector, and because the eigenvalues sum to zero this floor is strictly negative.

That negative floor is the whole problem. A region of the flow that is fully locked contributes at most 1 to the average, but compressed regions contribute negative values and can pay for it. Suppose a fraction w of the enstrophy is perfectly locked. Then $D = w + (1 - w) \cdot d_{\text{rest}}$, and the largest w consistent with an observed D is obtained by putting the remainder on the floor:

$$w_{\max} = (D - \lambda_3/\lambda_1) / (1 - \lambda_3/\lambda_1)$$

Evaluating this on the measured eigenvalue means of the July run gives Table 3.

t	D observed	$\langle \lambda_1 \rangle$	$\langle \lambda_3 \rangle$	pointwise floor	w_max
4.0	0.3478	0.5590	-0.6713	-1.2009	0.704
7.0	0.1973	0.7634	-0.8910	-1.1671	0.630
10.0	0.2543	0.7195	-0.8408	-1.1686	0.656

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Table 3. The largest locked-ensrophy fraction consistent with the observed global depletion ratio, at three times through the ensrophy peak.

At the ensrophy peak, where the July paper reports its strongest evidence, $D = 0.197$ is consistent with anything up to 63 per cent of the total ensrophy being perfectly aligned with the most-stretching eigenvector. The measured locked fraction there is 0.042 on the resolved grid and 0.048 on the coarse one. The observation therefore constrains alignment locking thirteen to fifteen times more loosely than the direct measurement of that locking, which appeared in the same table as a supporting column.

The consequence for the reading is immediate. The equivalence chain proposed in July — that global regularity is equivalent to the depletion ratio remaining bounded away from unity — is built on a variable that cannot detect a concentrating locked core, which is exactly the geometry a self-similar singularity exhibits. A global mean averages the core away. This is not a novel observation about fluid mechanics; it is the reason the Constantin–Fefferman criterion is stated pointwise. What is new is the quantification: on this flow, at this time, the blindness is a factor of thirteen.

The repair is available in the same data. The locked-ensrophy fraction L is local in the relevant sense — it counts where the dangerous configuration actually sits and cannot be subsidised by compressed regions elsewhere. The remainder of this paper treats L as the primary variable.

7. Reynolds scaling of the locked fraction

Table 4 collects every run performed for this paper, reporting the value of each diagnostic at that run's ensrophy peak, with the resolution parameter and its verdict.

Re	N	$k_{\max} \cdot \eta$	peak Ω	D	L	verdict
100	64	1.97	0.648	0.4881	0.2345	resolved
200	64	1.17	1.279	0.3571	0.0962	resolved
400	96	1.11	2.182	0.2023	0.0421	resolved
400	64	0.72	2.215	0.2538	0.0992	marginal
800	112	0.75	4.537	0.2133	0.0856	marginal
800	64	0.41	5.160	0.1665	0.1149	under-resolved
1600	64	0.24	11.146	0.0922	0.1129	under-resolved
3200	64	0.15	20.223	0.0483	0.1055	under-resolved

Table 4. All runs. Values are taken at the ensrophy peak of each run, meaning the largest value among that run's reported diagnostic samples rather than a temporally resolved maximum. The three bold rows are the only ones meeting the conventional resolution criterion.

Restricting attention to the three resolved rows, the locked fraction at the inherited cutoff of 0.90 falls monotonically with Reynolds number:

$$\text{Re} = 100 \rightarrow 200 : d \log L / d \log \text{Re} = -1.285$$

$$\text{Re} = 200 \rightarrow 400 : d \log L / d \log \text{Re} = -1.191$$

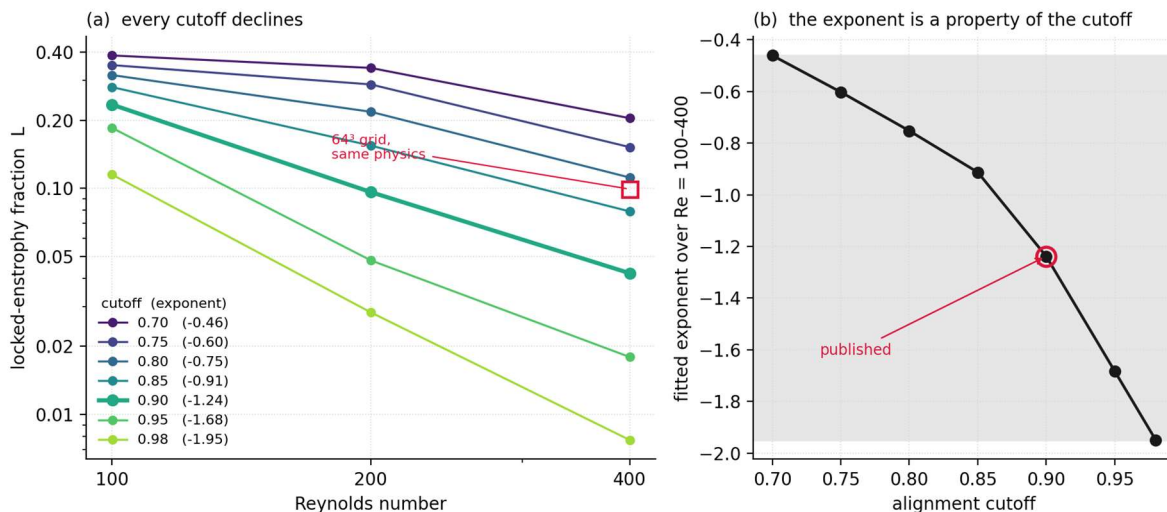
The two slopes agree to within seven per cent, and it is tempting to report $L \propto \text{Re}^{(-1.24)}$ as a measured scaling. That would be wrong, and the reason is the second confound of this paper.

7.1 The exponent is a property of the cutoff

The locked fraction is defined by an alignment cutoff, set at 0.90 in the July 2026 work and inherited here without examination. Recomputing the three resolved runs while accumulating the locked fraction at seven cutoffs between 0.70 and 0.98 gives Table 5.

cutoff	half-angle	L (Re=100)	L (Re=200)	L (Re=400)	decline 100→400	fitted exponent
0.70	45.6°	0.3868	0.3406	0.2042	1.89×	-0.461
0.75	41.4°	0.3506	0.2880	0.1521	2.31×	-0.602
0.80	36.9°	0.3167	0.2179	0.1116	2.84×	-0.752
0.85	31.8°	0.2802	0.1544	0.0790	3.55×	-0.913
0.90	25.8°	0.2345	0.0962	0.0421	5.57×	-1.238
0.95	18.2°	0.1849	0.0480	0.0179	10.33×	-1.682
0.98	11.5°	0.1150	0.0283	0.0077	14.94×	-1.949

Table 5. The locked-entropy fraction at seven alignment cutoffs, on the three resolved runs. The bold row is the inherited definition used in Table 4 and in the July 2026 work.



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Figure 1. (a) Locked-ensrophy fraction against Reynolds number on the three resolved runs, at seven alignment cutoffs; the heavy line is the inherited cutoff of 0.90. Every cutoff declines, and the separation between them widens with Reynolds number. The open red square is the $Re = 400$ value recomputed on the coarser 64^3 grid — the same physical state, displaced by roughly the distance a doubling of Reynolds number produces. (b) The fitted exponent over $Re = 100$ – 400 as a function of the cutoff. The shaded band spans the full range of exponents obtainable from the same three runs by moving the threshold; the circled point is the inherited definition that yields -1.24 .

Figure 1 shows both halves of this. The exponent varies from -0.461 to -1.949 , a range of 1.489 — wider than any single value within it, and wider than the published figure itself. The scaling exponent is therefore not a property of the flow. It is a property of where the cutoff is placed, and the value of -1.24 quoted from the 0.90 definition should not be cited as a measured exponent for the Navier–Stokes alignment geometry.

Two things survive this test, and the second is worth more than what it replaces.

The sign is robust. At every cutoff from 0.70 to 0.98 the locked fraction decreases monotonically with Reynolds number across the resolved range. Stated without extrapolation: over $Re = 100$ to 400 , the fraction of enstrophy in near-aligned configurations declines with Reynolds number, and that statement does not depend on where the alignment threshold is drawn.

The steepening is systematic, and it says something the single exponent concealed. Across $Re = 100$ to 400 the moderately aligned population at a cutoff of 0.70 thins by a factor of 1.89 , while the near-perfectly aligned tail at 0.98 thins by 14.94 . The tail depletes about eight times faster than the bulk. A distribution that were merely shifting toward lower alignment would deplete every cutoff by a comparable factor; a monotone steepening of this size means the alignment distribution is changing shape, with the extreme tail thinning fastest. That is a stronger and more specific statement about the geometry than any exponent, and it is what should be tested at higher Reynolds number.

Whether either behaviour continues above $Re = 400$ is unresolved, for the reason given in the next section.

8. Resolution error and signal are the same size

The sweep cannot be continued at fixed grid, and the reason is the most useful result in this paper.

At a single Reynolds number, $Re = 400$, the locked fraction at the enstrophy peak is 0.0421 on a 96^3 grid with $k_{\max} \cdot \eta = 1.11$, and 0.0992 on a 64^3 grid with $k_{\max} \cdot \eta = 0.72$. Nothing physical changed. The ratio is 2.36 .

The physical trend measured in the previous section is a factor of about 2.3 per doubling of Reynolds number. These two numbers are the same size. It follows that a sweep conducted at fixed grid resolution measures the grid at least as strongly as it measures the fluid, and any trend extracted from such a sweep is uninterpretable.

The under-resolved rows of Table 4 illustrate this directly. At 64^3 the locked fraction reads 0.1149 , 0.1129 and 0.1055 at $Re = 800$, 1600 and 3200 — an apparent plateau. That plateau sits essentially on top of the marginal $Re = 400$ value of 0.0992 obtained on the same grid. The grid is setting the number.

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Those three points are artefacts and are reported here only so that they are not mistaken for data elsewhere.

The marginal $Re = 800$ point at 112^3 is genuinely undecided. It gives $L = 0.0856$ at $k_{\max} \cdot \eta = 0.75$, which breaks the falling trend, but the known resolution bias at $Re = 400$ runs in exactly that direction and is more than large enough to account for the whole of it; a properly resolved value could plausibly lie anywhere between 0.03 and 0.09. It is not evidence for or against the scaling. A second defect compounds this. For reasons of computational cost that run was sampled at three times only — $t = 0$, 4.75 and 9.5 — so its reported peak is simply its final sample, and the true maximum was not located at all. Every other row in Table 4 was sampled at eleven times at unit spacing. The $Re = 800$ point at 112^3 is therefore doubly uninterpretable, on resolution and on temporal sampling, and it is retained in the table only for completeness.

Closing the range therefore requires resolution to grow with Reynolds number roughly as $Re^{(3/4)}$. Taking $k_{\max} \cdot \eta \geq 1.1$ as the standard, $Re = 800$ needs approximately $N = 160$ and $Re = 1600$ needs approximately $N = 256$. Both are ordinary workstation calculations — a 256^3 pseudospectral run stores on the order of three gigabytes of field data — and neither is feasible in the environment used here, where 112^3 already required 260 seconds of wall time to $t = 9.5$.

9. Push, pull, and the separation of the two halves

A structural observation clarifies why statements (C) and (D) fell while (A) and (B) did not, and it also identifies the only non-circular form of the argument this programme has been making.

Write the momentum equation and inspect what drives the fluid. Pressure enters as $-\nabla p$. Body forces enter as $-\nabla \Phi$. Viscosity is diffusion, which is transport down a gradient. The advective term $u \cdot \nabla u$ moves momentum but is not a force. Every native driving term is a gradient; the fluid falls, it is not pushed. The single exception is the external force f .

Statements (C) and (D) permit f . Statements (A) and (B) set it to zero. The half of the Clay formulation containing the only push in the system was resolved; the half consisting entirely of gradients was not touched. On this reading that is not a coincidence but a description of what was proved: a configuration was exhibited that a fluid cannot enter on its own and can be held in from outside, together with a demonstration that holding it requires only a smooth force.

For incompressible flow the pull is moreover global and instantaneous. Taking the divergence of the momentum equation gives a Poisson equation for the pressure, $\Delta p = -\rho \nabla \cdot (u \cdot \nabla u)$, whose solution at any point depends on the velocity field everywhere at that instant. A direct demonstration: a divergence-free velocity blob of Gaussian width 0.3 was placed at the centre of a 128^3 periodic box and the induced pressure computed spectrally.

radius R	max $ u $ beyond R	max $ p $ beyond R
0.5	6.89×10^{-1}	2.37×10^{-1}

radius R	max $ u $ beyond R	max $ p $ beyond R
1.0	2.19×10^{-2}	3.34×10^{-2}
1.5	3.36×10^{-5}	9.82×10^{-3}
2.0	2.87×10^{-9}	4.13×10^{-3}
2.5	1.46×10^{-14}	2.22×10^{-3}
3.0	4.65×10^{-21}	1.72×10^{-3}

Table 6. Fields as fractions of their own peak. The velocity is compactly supported to machine precision; the pressure it induces is not.

At radius 3.0 the velocity has fallen to 4.65×10^{-21} of its peak — it is gone — while the pressure is still at 1.72×10^{-3} . Eighteen orders of magnitude separate them. Nothing has propagated to the far field and nothing has moved there, and the far field is already being pulled, because the incompressibility constraint is solved globally at every instant. This is the same nonlocal singular structure that appears in the Biot–Savart representation of the strain in terms of the vorticity, and it is the reason the alignment map is self-referential: the vorticity computes, through a global integral, the strain that stretches it.

This also disposes of a circularity. An axiom stating that no terminal state is permitted cannot be invoked as proof that Navier–Stokes has no terminal state; that assumes the conclusion. The argument only has content if the continuation property is derived inside the fluid state space without importing it, and the gradient structure above is what such a derivation would rest on: a closed flow has no push available with which to hold a configuration it does not select. Whether that can be converted into a bound is unresolved and is addressed in the next section.

10. What this establishes and what it does not

Established. An independent engine reproduces the July 2026 Taylor–Green measurements to three or four significant figures, so the strain ledger, the intermediate-eigenvector wash and the production budget crossing are reproducible facts rather than artefacts of one implementation. The globally averaged depletion ratio is quantitatively blind to locked alignment, by a factor of thirteen at the enstrophy peak of this flow, and the locked-enstrophy fraction is the quantity that carries the constraint. Over the resolved range $Re = 100$ to 400 the locked fraction declines monotonically with Reynolds number at every alignment cutoff tested between 0.70 and 0.98, and the near-aligned tail declines about eight times faster than the moderately aligned bulk. At fixed grid, resolution error in this diagnostic is the same size as the physical Reynolds trend, so fixed-grid sweeps of it are uninterpretable.

Not established, and not claimed. Nothing here bears on global regularity. The measured scaling covers two intervals in Reynolds number, entirely below $Re = 400$, in one flow, from one initial condition, on a periodic domain. It is a trend, not a law, and it is not a bound. No uniform constant has been produced.

The behaviour above $Re = 400$ is unknown, not favourable — the one marginal point available breaks the trend and cannot be interpreted.

Falsified within this paper. The fitted exponent of -1.24 is not a measured property of the flow. Recomputing the same three runs across alignment cutoffs from 0.70 to 0.98 moves the exponent from -0.461 to -1.949 , a spread exceeding the value itself, so any single exponent quoted from this diagnostic is a statement about the threshold. This test was proposed as future work in an earlier draft of this paper and then performed before release; the result removed the headline number and replaced it with the sign and the shape change, both of which are cutoff-independent.

Retracted. The February 2026 claim that positive viscosity guarantees global regularity, and the theorem asserting smoothness for all time, are false and are withdrawn.

Superseded. The July 2026 equivalence chain, in which global regularity is stated as equivalent to the depletion ratio remaining bounded away from unity, is replaced by a statement in terms of the locked- ϵ fraction. The measurements that paper reported are unaffected; the variable in which its conclusion was drawn is.

Not a discovery. That averaged criteria cannot bound a concentrating core has been understood since the geometric-depletion programme of the early 1990s, which is why its criterion is pointwise. The contribution here is a number attached to that fact on a specific flow, and the substitution of a measurable local variable that a direct simulation already computes.

11. Next runs

Three calculations follow directly and are stated so that they can be executed or refuted by others.

First, complete the sweep at adequate resolution. $Re = 800$ at $N = 160$ and $Re = 1600$ at $N = 256$, holding $k_{\max} \cdot \eta$ above 1.1 at the enstrophy peak, extend the measured range to a factor of sixteen in Reynolds number. The prediction is that the locked fraction continues to fall with an exponent near -1.2 . If it flattens or rises, the reading advanced here is wrong, and that outcome is worth as much as the other.

Second, abandon the threshold entirely and measure the distribution. The cutoff family of Section 7.1 is a coarse sampling of a single object: the enstrophy-weighted probability density of the alignment cosine $|\cos(\omega, e_1)|$. The steepening result says that density is changing shape with Reynolds number rather than translating, and the density itself is no harder to compute than the seven cutoffs already reported. What should be tracked is its tail — for instance the exponent of its approach to unity, or its moments above some fixed reference — with the whole distribution reported at each Reynolds number so that no threshold need be chosen at all. A diagnostic that requires a free parameter invites the failure demonstrated in Section 7.1, and this one does not need one.

Third, evaluate the diagnostics on the September 2026 blowup construction rather than on decaying turbulence. That construction is explicit and its velocity field can be differentiated. The quantities to compute are the pointwise depletion in the contracting core, the locked fraction restricted to the core, and the contribution of the external force to the enstrophy budget. The prediction implied by Section 9

is specific: the core locks, and the forcing term accounts for the excess of production over dissipation. If the core locks and the force does not account for the excess, the gradient argument fails and should be abandoned.

A fourth item is named rather than proposed, because no route to it is offered here. The uniform bound remains open. Converting a measured decline in the locked fraction into a constant independent of time, Reynolds number and initial data — surviving the nonlocal singular integral that generates the strain from the vorticity — is the step that would matter, and nothing in this paper approaches it.

12. Prior art

The geometric-depletion programme, and in particular the pointwise vorticity-direction criterion, is due to Constantin and Fefferman (1993) and to Constantin, Fefferman and Majda. The preferential alignment of vorticity with the intermediate strain eigenvector was reported by Ashurst, Kerstein, Kerr and Gibson (1987). The blowup criterion in terms of the time integral of the vorticity supremum is due to Beale, Kato and Majda (1984). The Taylor–Green initial condition is due to Taylor and Green (1937). The official formulation of the Millennium problem, including the four statements and the admissibility conditions on the force, is due to Fefferman.

The construction technique used in the September 2026 result — building singularity by successive amplification of oscillatory layers while holding every derivative of the source bounded — was developed by Martínez-Zorúa and by Córdoba, with related extensions by Córdoba, Martínez-Zorúa and Zheng. The concurrent forced Euler result is due to Alpöge and Buckmaster. The Lean formalization and the multi-agent search that produced the Navier–Stokes application are OpenAI's.

The July 2026 measurements referenced throughout are from this programme's own earlier work and are cited as such, not as independent confirmation.

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The September 2026 items are announcements and preprints rather than refereed publications, and are cited as such. The Alpöge–Buckmaster preprint and the OpenAI writeup were consulted within two days of release; readers should check for revised versions, since the OpenAI document was itself revised on the day of announcement to expand its acknowledgement of the Córdoba–Martínez-Zoroa lineage.

A reproducibility package accompanies this paper. It contains the simulation engine, the twelve-check conservation and regression suite, cached diagnostic series for every run reported here, and scripts that regenerate Tables 1–6 and Figure 1 from either the cached data or a fresh computation. The full recomputation takes about five minutes on one core. All runs were performed in double precision. The reproduction reported in Section 5 was carried out before any Reynolds number other than 400 was attempted, and the resolution study of Section 8 was carried out after the sweep of Section 7 rather than before, which is why the under-resolved rows exist at all.